Roll No. | | | | | | | |

MID SEMESTER RE EXAMINATION 2025-26

Name of the Course: B.Tech in Petroleum Engineering

Semester: V

Name of the Paper: Machine Learning for Reservoir Characterization

Paper Code: TPE-511

Time: 1:30 Hrs

Maximum Marks: 50

Note: -

(i) All questions are compulsory.
(ii) Do any one of the (a) or (b) in each question
(iii) Each question carries 10 marks

Q1	(10marks)	**CO1,CO2**
(a)	Define Data Analytics and explain its types. Discuss how Exploratory Data Analysis (EDA) fits into the overall data analytics workflow. Why is EDA considered a critical step before applying machine learning models? Support your answer with real-world examples.	
	OR	
(b)	Explain Descriptive Statistics in the context of EDA. Discuss the role of measures such as mean, median, variance, skewness, and kurtosis in understanding dataset characteristics. Illustrate with an example dataset where these measures reveal hidden insights.	
Q2	(10marks)	
(a)	What is Data Pre-processing? Describe in detail the different techniques such as data cleaning, handling missing values, normalization, standardization, and encoding categorical variables. Also explain why improper pre-processing can negatively affect machine learning models.	**CO1, CO2**
	OR	
(b)	Visualization plays a key role in EDA. Compare different visualization methods such as histograms, scatter plots, box plots, pair plots, and heatmaps. Discuss which visualization is most useful for detecting outliers, correlations, and class separability in datasets with examples.	
Q3	(10marks)	**CO1, CO3**
(a)	Discuss the challenges in real-world data preparation such as data imbalance, missing data, noisy data, and high dimensionality. Suggest strategies to handle these challenges effectively during pre-processing.	
	OR	
(b)	Explain the working of a Decision Tree algorithm. Using the dataset below, manually construct a small decision tree using the ID3 algorithm based on Information Gain. Show entropy and gain calculations step by step.	

Weather	Temperature	Humidity	Windy	Play (Yes/No)
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rain	Mild	High	False	Yes
Rain	Cool	Normal	False	Yes
Rain	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No

Q4 (10 marks) **CO1, CO2, CO5**

(a) A dataset of house prices contains the following features:

Size (sq. ft.)	Price (lakhs)
1000	50
1500	65
2000	80
2500	95
3000	110

i.) Derive the regression equation.

ii.) Predict the price of a house with Size = 2200 sq. ft.

OR

(b) Explain Ensemble Learning methods with special reference to Bagging, Boosting, and Random Forests. Compare their working mechanisms, strengths, and weaknesses.

Q5 (10 marks) **CO1, CO2, CO5**

(a) What is Logistic Regression? Derive the logistic regression hypothesis and explain how the Sigmoid function transforms linear regression into a classification model. Using the dataset below of students' study hours vs. exam outcome, calculate the probability of passing when a student studies 3.5 hours.

Hours Studied	Pass (1)/Fail (0)
1	0
2	0
3	0
4	1
5	1
6	1

OR

(b) Explain Cluster Analysis. Compare K-Means clustering and Hierarchical clustering with suitable diagrams.

Given the following dataset of points:

Point	X	Y
P1	1	2
P2	2	3
P3	5	8
P4	6	9
P5	1	0

(i) Perform K-Means clustering with k=2, showing centroid updates step by step.

(ii) Perform Hierarchical clustering (single linkage) and show the dendrogram grouping order.